

JUSTICE-CHAIN SENTINEL · ENGINEERING NOTES

Sentinel Build Log

What changed today, working through the blueprint's build order on the `ipfs-changes` branch. Each section is marked by how far it's actually been proven, not just written.

Foundation fixes

Blueprint Phase 1 · live-verified on a physical device

Verified on device

- **Mesh node-ID bug** — `mesh_service.dart` was reading a Hive key nothing ever wrote, so every device broadcast as `UNKNOWN_NODE`. Now reads from `IdentityService.getMyNodeId()`.
- **Hive records keyed by content hash**, not insertion position — the old `getAt` / `putAt` pattern silently drifted whenever a record was added or removed.
- **45-second failsafe moved into `EmergencyController`**, off the screen's widget lifecycle — it used to stop working the moment the recording screen was disposed or the phone locked. Confirmed firing at 45.00–45.05s on-device, independent of the UI.
- **PIN hashing** moved from plain SHA-256 to Argon2id with a random salt.
- **Debug affordances** (the IPFS test button, simulate-distress, print-vault) gated behind `kDebugMode` so they can't ship in a release build.
- **AI trigger hardened**: now requires 2 of the last 3 one-second windows to clear an 0.80 confidence threshold, instead of firing on any single noisy window. This is what stopped a live retrigger loop that was crashing the camera mid-test.

Core evidence encryption

Blueprint Phase 2 (encryption half) · live-verified, real CID on IPFS

Verified on device

Every recording is now AES-256-GCM encrypted with a random per-incident key before it touches disk or the network. Previously the app uploaded raw `.mp4` plaintext straight to public IPFS.

- New `EvidenceEncryptor` (`lib/core/evidence_encryptor.dart`): encrypts, wraps the incident key to the device's own vault key, SHA-256 hashes and Ed25519-signs the plaintext, then deletes the plaintext — before upload even starts.

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- EvidenceVaultService now uploads the .enc ciphertext to Pinata, not the original clip.
- Confirmed on-device: a fresh recording leaves only a .enc file behind (no matching .mp4), and the ciphertext uploads to a real CID.

Guardian key sharing

Blueprint Phase 2 (guardian half) · unit-tested only, no second device available to confirm live

Code complete, not device-tested

A guardian's phone should be able to decrypt a clip without ever holding the sender's private key. Implemented as a standard sealed box: X25519 key agreement → HKDF → AES-GCM.

- New X25519 keypair generated per device, with a migration path so an already-registered device gets one on its next launch without disturbing its existing identity.
- QR payload (IdentityService.getQrPayload) now carries the X25519 public key; GuardianManager stores it per guardian.
- EvidenceEncryptor.encryptAndSeal wraps the incident key a second time for every currently-paired guardian.
- Verified with two independent in-process keypairs (unit tests) that a guardian can unwrap and decrypt using only their own key and the sender's public key — and that a mismatched key fails loudly (GCM auth failure).

Known gap, not an oversight: any one paired guardian can currently decrypt alone. The blueprint's finished design wants a 2-of-3 Shamir quorum so no single guardian (who could be the attacker) can watch a clip unilaterally — that upgrade, plus a retry/outbox for delivering key-shares and issuing them to guardians paired after the fact, is still open.

Self-view playback screen

New screen, not in the original blueprint phases · built, live-check pending

Built, not yet confirmed on device

Nothing in the app could open a .enc file at all until this. Added EvidenceViewerScreen (lists your own vault records) and EvidencePlayerScreen (decrypts to a temp file, plays it, deletes the temp file on close). Reachable from the home screen via **My Evidence**.

Local blockchain anchoring

Blueprint Phase 3 · live-verified end to end, on-chain event checked by hand

Verified on device

Rewrote `contracts/JusticeLedger.sol` to the blueprint's v2 interface — `storeEvidence` / `verifyEvidence` / `getEvidenceHistory` , self-authenticated by recovering the signer's address with `ecrecover` rather than trusting whoever submits the transaction.

- Each device now also holds a `secp256k1` anchoring keypair, used only to sign manifest hashes (never to pay gas).
- New `AnchoringService` (`lib/core/anchoring_service.dart`) submits the signed anchor against a local Hardhat node after a successful IPFS upload, using a funded Hardhat test account as the gas-paying submitter.
- 5 new Hardhat tests (`test-contracts/JusticeLedger.test.js`) cover storage, duplicate rejection, lookup-by-hash, and multi-record history.
- Confirmed by reading the chain's raw event logs directly: the on-chain `EvidenceAnchored` event's CID matched the IPFS upload's CID exactly.

Bug found and fixed today: the first anchoring attempt failed with “signed for another chain” — the client wasn't specifying a chain ID, so `web3dart` defaulted to mainnet (1) instead of Hardhat's local chain (31337). Fixed by fetching the chain ID from the node instead of hardcoding one, which also means it won't need touching again for the public-testnet step.

Environmental gotcha, not a code bug: uploads hung indefinitely on one Wi-Fi network (confirmed by uploading the same file successfully from a dev machine and from the phone on mobile data). `PinataService` 's HTTP call has no timeout, so a network silently killing large POST requests looks identical to a hang. Worth a timeout + clearer error surfacing as a follow-up, independent of anything above.

What's left

PHASE	STATUS
Phase 2 — Hive outbox, retroactive guardian keys, Shamir 2-of-3 quorum	Not started
Phase 4 — Mesh relay (manifest-first transfer, advertising/discovery never wired up, Android 11+ URI handling)	Not started
Phase 5 — Guardian playback & on-chain verification screen	Not started

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Phase 6 — 15-second capture segments, Android foreground service Not started

Phase 7 — Public testnet (Polygon Amoy) Not started

Phase 8 — Hyperledger Fabric Parked — gated on a real police/lab/court partner, per the blueprint itself